Payoff families and value in games with one-sided information

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Abstract

We study bounded Borel payoffs in repeated zero-sum games with a finite hidden state known to one player. For fixed finite action sets, value existence at every prior is equivalent to a condition on posterior and capital arrays over the public history tree. The infimum of the initial capital excess equals the minimax gap. These certificates yield an exact cover of the value-admitting payoffs by Borel selections of pathwise closed convex regions. Each family has a greatest payoff of Baire class at most three, a common value at every prior, and closure under public-path Borel mixing. Independently of value existence, every bounded Borel payoff has a majorant of Baire class at most three with the same upper value at every prior. The proofs use Borel superhedging duality on finite-alphabet trees and an explicit posterior reduction. All pathwise conditions include plays of probability zero.

1 Model and question

Let K, I, J be nonempty finite sets with the discrete topology. Write $\mathbb{N} = \{0, 1, \ldots\}$ and

$$X = I \times J, \qquad \mathcal{H} = \bigcup_{t \geq 0} X^t, \qquad \Omega = X^{\mathbb{N}}.$$

Equip Ω with the product topology, with X discrete. A history $h \in \mathcal{H}$ is public; $\varnothing$ denotes the empty history, hij appends (i, j) to h, and $\omega|t$ is the prefix of length t.

Nature draws $k \in K$ with prior $p \in \Delta(K)$ and reveals it only to the maximizing player. At every date the players choose actions simultaneously and then both observe the pair. Their behavioral strategies are

$$\sigma : K \times \mathcal{H} \longrightarrow \Delta(I), \qquad \tau : \mathcal{H} \longrightarrow \Delta(J).$$

The two current actions are drawn independently conditional on the state and history. For a bounded Borel payoff $f : K \times \Omega \to \mathbb{R}$, let $\gamma_{p,f}(\sigma, \tau) = \mathbb{E}_{p,\sigma,\tau} f(k, \omega)$ and

$$\underline{v}_f(p) = \sup_{\sigma} \inf_{\tau} \gamma_{p,f}(\sigma, \tau), \qquad \overline{v}_f(p) = \inf_{\tau} \sup_{\sigma} \gamma_{p,f}(\sigma, \tau).$$

The game has a value at p when these two quantities are equal; their common value is then denoted by $v_f(p)$.

Castellon Koltun, Lehrer and Solan [1, Section 5] ask for properties of the payoff that guarantee value existence in this model. The prior-uniform formulation in problem C73-8 of [2] asks for a characterization of

$$\mathcal{V}_{K,I,J} = \{f : K \times \Omega \to \mathbb{R} \text{ bounded Borel} : \underline{v}_f(p) = \overline{v}_f(p) \text{ for every } p \in \Delta(K)\}.$$

We express conditions on the payoff and the information structure through local convex geometry and pointwise boundary inequalities. The resulting arrays determine payoff families and

describe path-dependent changes within each family that preserve the value at every prior. The characterization is infinitary; no effective recognition procedure for an arbitrary Borel payoff is asserted.

Choose $a < b$ with $a \le f \le b$ and put $F = (f-a)/(b-a)$. Thus $0 \le F \le 1$. Both values transform by this positive affine map. We write $\mathcal{V}^{[0,1]}_{K,I,J}$ for the normalized class and $\Delta_{\mathbb{Q}}(K)$ for the rational-coordinate priors. The strategy spaces do not depend on p, including at its boundary. Consequently,

$$|\underline{v}_F(p) - \underline{v}_F(p')| \le \|p - p'\|_1, \qquad |\overline{v}_F(p) - \overline{v}_F(p')| \le \|p - p'\|_1. \tag{1}$$

These bounds hold for every fixed strategy pair and survive either order of infimum and supremum. Both values are also monotone in F.

2 Local capital certificates

Definition 2.1 (Capital arrays at prior p)**.** At each $h \in \mathcal{H}$, assign

$$q(h) \in \Delta(K), \quad A_k(h), B(h) \in [0,1] \ (k \in K).$$

Require $q(\varnothing) = p$ and the following conditions.

Posterior children:

$$q(hij) \text{ is independent of } j; \quad \text{denote it by } \bar{q}_i(h). \tag{BF}$$

In particular, a posterior child depends on the current informed action but is independent of the current uninformed action, including branches that will receive zero convex weight.

Common-column capital geometry: define

$$V_j(h) = \big(A_k(hij) - A_k(h)\big)_{(k,i)\in K\times I} \in \mathbb{R}^{K\times I}.$$

Require

$$\operatorname{conv}\{V_j(h) : j \in J\} \cap \mathbb{R}^{K\times I}_{-} \neq \varnothing, \tag{UC}$$

where $\mathbb{R}^{K\times I}_{-}$ is the coordinatewise nonpositive orthant.

Posterior and lower-capital geometry: define

$$Z_i(h) = \big(\bar{q}_i(h), (B(hij) - B(h))_{j\in J}\big) \in \mathbb{R}^K \times \mathbb{R}^J.$$

Require

$$\operatorname{conv}\{Z_i(h) : i \in I\} \cap \big(\{q(h)\} \times \mathbb{R}^J_{-}\big) \neq \varnothing. \tag{LC}$$

Define the bounded Borel function on *every* path

$$G_q(\omega) = \liminf_{t\to\infty} \sum_{k\in K} q_k(\omega|t) F(k,\omega). \tag{G}$$

Require the pointwise boundary conditions

$$\liminf_{t\to\infty} A_k(\omega|t) \ge F(k,\omega) \qquad (k \in K, \omega \in \Omega), \tag{UB}$$

$$\liminf_{t\to\infty} B(\omega|t) \ge 1 - G_q(\omega) \qquad (\omega \in \Omega). \tag{LB}$$

These include paths of probability zero.

Definition 2.2 (Payoff certificate condition)**.** The payoff F satisfies condition (C) if for every $p \in \Delta_{\mathbb{Q}}(K)$ and every integer $n \ge 1$, there are capital arrays as above with

$$\sum_{k\in K} p_k A_k(\varnothing) + B(\varnothing) \le 1 + \frac{1}{n}. \tag{RC}$$

The local conditions are finite convex-hull intersections at each history. Their boundary inequalities compare the arrays with the payoff on every path. We call arrays satisfying these requirements a capital certificate; the root bound is imposed separately in (C).

Theorem 2.3 (Certificate characterization)**.** *For arbitrary fixed nonempty finite K, I, J and every bounded Borel f,*

$$f \in \mathcal{V}_{K,I,J} \quad \Longleftrightarrow \quad F \text{ satisfies (C)}.$$

The equivalence is independent of the normalization bounds a, b.

The same certificates give an exact representation of the value gap. Let $D_F(p)$ be the infimum of $\sum_k p_k A_k(\varnothing) + B(\varnothing) - 1$ over capital arrays at prior p satisfying (BF)–(LB), without (RC). Such arrays always exist: take $q(h) = p$ and $A_k(h) = B(h) = 1$.

Theorem 2.4 (Exact gap representation)**.** *For every bounded Borel $F : K \times \Omega \to [0,1]$ and every prior p,*

$$D_F(p) = \overline{v}_F(p) - \underline{v}_F(p).$$

3 Proof of the capital representation

3.1 Borel superhedging on a finite-alphabet tree

We use equation (3) and Theorems 2 and 13 of T'Joens and De Bock [3]. Equation (3) defines the game-theoretic upper expectation by pointwise superhedging. Theorem 2 identifies it with the axiomatic upper expectation, and Theorem 13 identifies the latter with the measure-theoretic upper expectation for bounded Borel payoffs. Together they give the following form:

Lemma 3.1 (Borel superhedging)**.** *On a finite alphabet tree, assign each history h a nonempty closed convex set P_h of next-step probability vectors. For a bounded nonnegative Borel W, let c be the supremum of $\mathbb{E}_P W$ over all laws whose one-step kernels lie in P_h. For every $\delta > 0$ there is a real-valued, globally bounded-below supermartingale M with $M(\varnothing) < c + \delta$,*

$$\sup_{\pi \in P_h} \sum_{z \in X} \pi_z M(hz) \le M(h), \qquad \liminf_t M(\omega|t) \ge W(\omega) \quad (\omega \in \Omega).$$

The local upper expectation is $Q_h(Z) = \sup_{\pi \in P_h} \pi \cdot Z$, so the probability sets and upper expectations agree in the terminology of [3]. Both applications below use bounded Borel payoffs and nonempty closed convex local models on the finite alphabet X.

Lemma 3.2 (Capital can be bounded by one)**.** *If $0 \le W \le 1$, the process in Lemma 3.1 may be chosen in $[0,1]$, with the same bound on its initial capital.*

Proof. Any process furnished by the lemma is nonnegative at every node. Its local inequality implies $\min_z M(hz) \le M(h)$. If $M(h) < 0$ at some node, successively choose a child whose value is no larger than its parent's. The resulting infinite path extending h has $\liminf M \le M(h) < 0$, contradicting its nonnegative boundary lower bound. A finite prefix does not affect this liminf.

Now put $\widetilde{M}(h) = \min\{M(h), 1\}$. For each $\pi \in P_h$, concavity and monotonicity of $z \mapsto \min\{z, 1\}$ give

$$\sum_z \pi_z \min\{M(hz), 1\} \le \min\left\{ \sum_z \pi_z M(hz), 1 \right\} \le \min\{M(h), 1\}.$$

Thus $\widetilde{M}$ satisfies the local inequality. Also $\liminf \widetilde{M} = \min\{\liminf M, 1\} \ge W$ on every path. Clipping decreases initial capital, and $\widetilde{M} \in [0,1]$. □

3.2 Posterior flow

Lemma 3.3 (Bayes-flow correspondence)**.** *Fix $p \in \Delta(K)$. Let $q : \mathcal{H} \to \Delta(K)$ satisfy $q(\varnothing) = p$ and* (BF)*, and let $w_i(h) \geq 0$ be coefficients summing to one with $q(h) = \sum_i w_i(h)\bar{q}_i(h)$. Define*

$$x_k(i \mid h) = \frac{w_i(h)\bar{q}_{i,k}(h)}{q_k(h)} \quad \text{when } q_k(h) > 0,$$

and use any fixed action distribution when $q_k(h) = 0$. This is an informed behavioral strategy. Against every uninformed strategy τ, the public posterior at every positive-probability history is $q(h)$, the informed action has public distribution $w(h)$, and

$$\mathbb{E}_{p,x,\tau} F(k,\omega) = \mathbb{E}_{p,x,\tau} G_q(\omega). \tag{PI}$$

Conversely every informed strategy x at prior p gives such arrays with these public laws.

Proof. At a positive-$q_k(h)$ row, the barycentric identity gives $\sum_i x_k(i \mid h) = 1$. If $q_k(h) = 0$, nonnegativity and that same identity imply $w_i(h)\bar{q}_{i,k}(h) = 0$ for every i. Hence

$$\sum_k q_k(h) x_k(i \mid h) = \sum_k w_i(h)\bar{q}_{i,k}(h) = w_i(h).$$

Induct on the length of a positive-probability public history h. At the root the posterior is $p = q(\varnothing)$. If the posterior at h is $q(h)$, the next pair (i,j) has conditional probability $w_i(h)\tau_j(h)$. When this is positive, Bayes' rule gives the next posterior

$$\frac{q_k(h)x_k(i \mid h)}{w_i(h)} = \bar{q}_{i,k}(h) = q_k(hij).$$

The current factor $\tau_j(h)$ cancels because the uninformed player's randomization does not depend on the hidden state. A zero-probability child's assigned posterior does not affect the induction or law. Zero-prior states are covered by the same argument.

The vector $q(\omega|t)$ is consequently the bounded posterior martingale under every τ. Its coordinates converge almost surely to the conditional probabilities of the hidden state given the entire public play. This is the upward conditional-expectation martingale convergence theorem [4, Theorem 4.6.8], because the finite public-history sigma fields generate the sigma field of the full public play. On that event,

$$G_q(\omega) = \sum_k q_{\infty,k}(\omega) F(k,\omega).$$

Condition each $\mathbf{1}_{\{k=\ell\}} F(\ell,\omega)$ on the entire public play, then sum over ℓ, to obtain (PI).

The definition of G_q also covers paths on which the posterior fails to converge. For each t, the function $\sum_k q_k(\omega|t)F(k,\omega)$ is bounded Borel. Taking the pointwise liminf over t therefore gives a bounded Borel function on every path.

Conversely start with any informed strategy x and $q(\varnothing) = p$. At each history put

$$w_i(h) = \sum_k q_k(h) x_k(i \mid h).$$

For $w_i(h) > 0$ put $\bar{q}_{i,k}(h) = q_k(h)x_k(i \mid h)/w_i(h)$, and for $w_i(h) = 0$ put $\bar{q}_i(h) = p$. Set $q(hij) = \bar{q}_i(h)$ for every j. These arrays satisfy (BF) and the barycentric identity. At a zero-weight branch all terms $q_k(h)x_k(i \mid h)$ are zero, so the chosen vector contributes nothing to the barycentric identity. The preceding induction proves the claimed public laws and (PI). □

3.3 The guarantees

Fix capital arrays at prior p. Condition (UC) provides at every h coefficients $y_j(h) \geq 0$, summing to one, such that

$$\sum_j y_j(h) A_k(hij) \leq A_k(h) \quad \text{for every } (k,i) \in K \times I.$$

These coefficients define an uninformed behavioral strategy y; the same distribution works for all states and all current informed actions, as required by its information restriction.

In each state k, against every informed strategy σ_k, the bounded nonnegative process $A_k(\omega|t)$ is a supermartingale. Indeed, condition on h and a pure informed action i, apply the displayed inequality, and then average over that informed action. Fatou's lemma and (UB) imply

$$\mathbb{E}_{k,\sigma_k,y} F(k,\omega) \leq \mathbb{E} \liminf_t A_k(\omega|t) \leq \liminf_t \mathbb{E} A_k(\omega|t) \leq A_k(\varnothing).$$

After weighting by p this proves

$$\overline{v}_F(p) \leq \sum_k p_k A_k(\varnothing). \tag{2}$$

Condition (LC) supplies convex coefficients $w_i(h) \geq 0$ summing to one, simultaneously satisfying

$$q(h) = \sum_i w_i(h) \bar{q}_i(h), \qquad \sum_i w_i(h) B(hij) \leq B(h) \quad (j \in J).$$

Reconstruct x using these coefficients and Lemma 3.3. Against every τ, the public next-action kernel is $w_i(h)\tau_j(h)$. Condition (LC) makes $B(\omega|t)$ a bounded nonnegative supermartingale: apply its inequality for each pure j and average over $\tau(h)$. Fatou, (LB), and (PI) give

$$\mathbb{E}_{p,x,\tau}(1-F) = \mathbb{E}(1-G_q) \leq \mathbb{E} \liminf_t B(\omega|t) \leq B(\varnothing).$$

Consequently

$$\underline{v}_F(p) \geq 1 - B(\varnothing). \tag{3}$$

The elementary minimax inequality, (2), and (3) yield

$$0 \leq \overline{v}_F(p) - \underline{v}_F(p) \leq \sum_k p_k A_k(\varnothing) + B(\varnothing) - 1. \tag{4}$$

Thus (C) gives equality at every rational prior by letting n tend to infinity.

The Lipschitz bounds (1) extend equality from the rational priors to the entire simplex. Positive affine normalization gives sufficiency for the original payoff.

3.4 Sharpness of the bounds

For arbitrary F and p write $L = \underline{v}_F(p)$ and $U = \overline{v}_F(p)$. Given $\delta > 0$, the definitions of the extrema provide behavioral strategies x, y such that

$$\inf_\tau \mathbb{E}_{p,x,\tau} F \geq L - \delta, \qquad \sup_\sigma \mathbb{E}_{p,\sigma,y} F \leq U + \delta. \tag{5}$$

No optimal strategy is assumed to exist. Obtain $q, w, \bar{q}$ from x using Lemma 3.3.

For each state k use the local credal sets

$$P_h^y = \{(a_i y_j(h))_{ij} : a \in \Delta(I)\}.$$

These are nonempty closed convex sets, being linear images of the finite simplex. Their local upper expectation of a next-step array Z_{ij} is

$$\max_{i \in I} \sum_j y_j(h) Z_{ij}.$$

Compatible precise kernels are exactly the kernels of arbitrary state-k informed behavioral responses to the fixed y: the marginal on I recovers $a(h)$. Therefore the compatible laws are exactly those response laws. Let c_k be their upper expectation of $F(k,\cdot)$. Lemma 3.1, with the finite alphabet X and this bounded Borel payoff, followed by Lemma 3.2, gives $A_k \in [0,1]$ with

$$A_k(\varnothing) < c_k + \delta, \qquad \sum_j y_j(h) A_k(hij) \le A_k(h) \quad (h \in \mathcal{H}, i \in I),$$

and with (UB) on every path. The same y supplies convex coefficients for all k, proving (UC) for the combined arrays.

The finitely many state components can be chosen independently, so

$$\sum_k p_k c_k = \sup_\sigma \mathbb{E}_{p,\sigma,y} F. \tag{6}$$

For every σ, $\mathbb{E}_{p,\sigma,y} F \le \sum_k p_k c_k$, so its supremum is also at most that sum. Conversely, choose ε-optimal responses separately in each positive-prior state, combine them into one informed behavioral strategy, and let $\varepsilon \downarrow 0$. Zero-prior states have no effect. Hence (5) gives

$$\sum_k p_k A_k(\varnothing) < U + 2\delta. \tag{7}$$

For the other capital process, use the public-law credal sets

$$P_h^w = \{(w_i(h) b_j)_{ij} : b \in \Delta(J)\}.$$

They are again nonempty closed convex linear images of a finite simplex. Their local upper expectation is $\max_j \sum_i w_i(h) Z_{ij}$. Compatible precise trees correspond exactly to uninformed strategies against x, because their J marginal is b and their public I marginal is w. This statement is also valid at histories of probability zero, whose specified kernels do not affect the root law.

The boundary payoff $W = 1 - G_q$ is bounded, nonnegative, and Borel. Lemma 3.3 shows that its upper expectation over these compatible trees equals

$$\sup_\tau \mathbb{E}(1 - G_q) = 1 - \inf_\tau \mathbb{E}_{p,x,\tau} F.$$

Apply Lemmas 3.1 and 3.2. They give $B \in [0,1]$ with $\sum_i w_i(h) B(hij) \le B(h)$ for every j, and satisfying (LB). Together with the barycentric identity, these inequalities show that $\sum_i w_i(h) Z_i(h)$ belongs to $\{q(h)\} \times \mathbb{R}^J_-$, proving (LC). The root bound is

$$B(\varnothing) < 1 - \inf_\tau \mathbb{E}_{p,x,\tau} F + \delta \le 1 - L + 2\delta. \tag{8}$$

Thus both local models and their boundary payoffs satisfy the hypotheses of Lemma 3.1.

Combining (7) and (8) yields capital arrays with

$$\sum_k p_k A_k(\varnothing) + B(\varnothing) - 1 < U - L + 4\delta.$$

Taking the infimum of these costs and letting $\delta \downarrow 0$, then using (4), proves $D_F(p) = U - L$. This establishes Theorem 2.4 for every bounded Borel payoff and every prior, without a value-existence assumption. When the game has a value, $U = L$; choosing $4\delta < 1/n$ gives (RC) for every n. This proves necessity of (C), and completes Theorem 2.3. Its equivalence with value existence also proves independence of the normalization bounds.

4 Upper-value regularization

The upper part of the proof gives a separate representation that does not use posterior arrays. Let

$$\mathcal{U}(F) = \{A : \mathcal{H} \to [0,1]^K : \text{(UC) and (UB) hold}\}.$$

This set has no designated prior.

Proposition 4.1. *For every normalized Borel payoff F and every prior p,*

$$\overline{v}_F(p) = \inf_{A \in \mathcal{U}(F)} p \cdot A(\varnothing). \tag{9}$$

Proof. The argument for (2) uses only (UC) and (UB), so it gives one inequality. The construction leading to (7) provides $A \in \mathcal{U}(F)$ with $p \cdot A(\varnothing) < \overline{v}_F(p) + 2\delta$ for every $\delta > 0$, proving the other. □

We use the convention that Baire class zero consists of continuous real-valued functions, and class at most $m+1$ consists of pointwise limits of sequences of functions of class at most m.

Lemma 4.2. *For a nonempty countable family of arrays $A^d : \mathcal{H} \to [0,1]^K$, the function on $K \times \Omega$ defined by*

$$c(k,\omega) = \inf_d \liminf_t A^d_k(\omega|t)$$

has Baire class at most three.

Proof. For fixed d, k, t, the prefix function $A^d_k(\omega|t)$ is continuous. For each N, its tail infimum is the pointwise limit, as $M \to \infty$, of the continuous functions $\min_{N \le t \le M} A^d_k(\omega|t)$. It has class at most one. The increasing limit over N is $\liminf_t A^d_k(\omega|t)$, of class at most two. Enumerate the indices d, repeating terms if the family is finite. The infimum over d is the decreasing limit of finite minima of these class-two functions, hence has class at most three. Finite minima preserve each class, by applying the continuous minimum operation to the defining sequences. The finitely many state coordinates can be combined without raising the class. □

Theorem 4.3 (Upper-value majorant)**.** *For every Borel $F : K \times \Omega \to [0,1]$ there is a function $C : K \times \Omega \to [0,1]$ of Baire class at most three such that*

$$F \le C, \qquad \overline{v}_C(p) = \overline{v}_F(p) \quad (p \in \Delta(K)).$$

Every Borel G with $F \le G \le C$ has the same upper value as F at every prior.

Proof. For every $d = (r, n)$ in the countable set

$$\mathcal{D} = \Delta_{\mathbb{Q}}(K) \times \mathbb{N}_{\ge 1},$$

choose $A^d \in \mathcal{U}(F)$ with $r \cdot A^d(\varnothing) \le \overline{v}_F(r) + 1/n$, using (9). Set $C(k,\omega) = \inf_d \liminf_t A^d_k(\omega|t)$. Then $F \le C$, Lemma 4.2 applies, and every A^d also belongs to $\mathcal{U}(C)$. For rational r and every n,

$$\overline{v}_F(r) \le \overline{v}_C(r) \le r \cdot A^{(r,n)}(\varnothing) \le \overline{v}_F(r) + 1/n.$$

Let $n \to \infty$, then use (1) to extend equality to all priors. The assertion for G follows by monotonicity. □

For an arbitrary bounded Borel f, apply the theorem to $F = (f-a)/(b-a)$ and set $\widehat{f} = a + (b-a)C$. Then $f \le \widehat{f}$ and $\overline{v}_{\widehat{f}}(p) = \overline{v}_f(p)$ for every prior, whether or not the game has a value.

Corollary 4.4. *Every $f \in \mathcal{V}_{K,I,J}$ has a bounded majorant $\widehat{f}$ of Baire class at most three such that every bounded Borel g with $f \le g \le \widehat{f}$ belongs to $\mathcal{V}_{K,I,J}$ and satisfies $v_g(p) = v_f(p)$ for every prior.*

Proof. Normalize f and apply Theorem 4.3. For $F \le G \le C$,

$$v_F(p) = \underline{v}_F(p) \le \underline{v}_G(p) \le \overline{v}_G(p) = \overline{v}_F(p) = v_F(p).$$

Undo the affine normalization, setting $\widehat{f} = a + (b-a)C$. □

The majorant theorem preserves the upper value even for games without a value. It is not a value-existence criterion. No optimality of the bound on the Baire class is asserted.

5 An exact cover by payoff families

We now keep full certificates rather than only their upper arrays. This yields families of payoffs with the same value at every prior. Their definition uses the information and action structure, without first choosing a payoff.

5.1 Boundary regions

An array family Θ assigns (q^d, A^d, B^d) to every $d = (p, n) \in \mathcal{D}$. Require $q^d(\varnothing) = p$, (BF)–(LC), and (RC) for each d. The payoff-dependent boundary conditions have not yet been imposed. Define

$$u_k^d(\omega) = \liminf_t A_k^d(\omega|t), \qquad c_{\Theta,k}(\omega) = \inf_{d\in\mathcal{D}} u_k^d(\omega),$$
$$\ell^d(\omega) = 1 - \liminf_t B^d(\omega|t), \qquad Q^d(\omega) = \bigcap_{N\ge 0} \overline{\{q^d(\omega|t) : t \ge N\}}.$$

The set $Q^d(\omega)$ is the nonempty compact cluster set in $\Delta(K)$, defined even when the posterior sequence does not converge. Impose the boundary compatibility condition

$$\min_{r\in Q^d(\omega)} r \cdot c_\Theta(\omega) \ge \ell^d(\omega) \qquad (d \in \mathcal{D},\ \omega \in \Omega). \tag{BC}$$

Let $\mathfrak{T}_{K,I,J}$ be the class of families satisfying these requirements. All arrays take values in the same simplexes and intervals as the capital arrays in Section 2. We identify c_Θ with the payoff $(k, \omega) \mapsto c_{\Theta,k}(\omega)$.

For $\Theta \in \mathfrak{T}_{K,I,J}$ define

$$\mathcal{P}_\Theta(\omega) = \left\{ z \in [0,1]^K : \begin{array}{ll} z_k \le c_{\Theta,k}(\omega) & (k \in K), \\ r \cdot z \ge \ell^d(\omega) & (d \in \mathcal{D},\ r \in Q^d(\omega)) \end{array} \right\}, \tag{10}$$

and let $\mathcal{S}_\Theta$ consist of the Borel payoffs F with $F(\cdot, \omega) \in \mathcal{P}_\Theta(\omega)$ at every path. The regions are intersections of closed half-spaces and the cube; they need not be polyhedra with finitely many defining inequalities.

Theorem 5.1 (Payoff-family representation)**.** *For arbitrary fixed nonempty finite K, I, J,*

$$\mathcal{V}_{K,I,J}^{[0,1]} = \bigcup_{\Theta\in\mathfrak{T}_{K,I,J}} \mathcal{S}_\Theta. \tag{11}$$

For each $\Theta \in \mathfrak{T}_{K,I,J}$ the following hold.

(i) *Each $\mathcal{P}_\Theta(\omega)$ is nonempty, closed and convex, and the correspondence $\omega \mapsto \mathcal{P}_\Theta(\omega)$ has a Borel graph. The greatest element of $\mathcal{S}_\Theta$ in the pointwise order is c_Θ, of Baire class at most three.*

(ii) *Every $F \in \mathcal{S}_\Theta$ has the same value function*

$$v_\Theta(p) = \inf_{d \in \mathcal{D}} p \cdot A^d(\varnothing) \qquad (p \in \Delta(K)). \tag{12}$$

In particular, v_Θ is concave and 1-Lipschitz in the ℓ^1 norm.

(iii) *If $F^1, \dots, F^m \in \mathcal{S}_\Theta$ and $\lambda : \Omega \to \Delta(\{1, \dots, m\})$ is Borel, then*

$$F^\lambda(k, \omega) = \sum_{a=1}^{m} \lambda_a(\omega) F^a(k, \omega) \quad \textit{belongs to } \mathcal{S}_\Theta.$$

Also, if $F \in \mathcal{S}_\Theta$ and G is Borel with $F \le G \le c_\Theta$, then $G \in \mathcal{S}_\Theta$.

The cover in (11) *is not asserted to be disjoint.*

5.2 Proof of the representation

For a sequence $q_t \in \Delta(K)$ with cluster set Q and any fixed $z \in \mathbb{R}^K$, compactness gives

$$\liminf_t q_t \cdot z = \min_{r \in Q} r \cdot z. \tag{13}$$

Every cluster point is the limit of a subsequence, so its scalar product is at least the liminf. Conversely, a subsequence realizing the scalar liminf has a further convergent subsequence in $\Delta(K)$, which attains equality. Apply this identity at each path, with $z = F(\cdot, \omega)$. The lower boundary inequalities in (10) are exactly (LB) for all d, while $F \le c_\Theta$ is exactly (UB) for all d. Thus, for Borel $F : K \times \Omega \to [0, 1]$,

$$F \in \mathcal{S}_\Theta \quad \Longleftrightarrow \quad (q^d, A^d, B^d) \text{ certifies } F \text{ at its prior for every } d. \tag{14}$$

Condition (BC) says that $c_\Theta(\omega)$ itself belongs to $\mathcal{P}_\Theta(\omega)$. Conversely, any point z of that region implies (BC), since $z \le c_\Theta$ and every $r \in \Delta(K)$ has nonnegative coordinates. This proves nonemptiness; closedness and convexity follow from (10). Lemma 4.2 shows that c_Θ is Borel and has the claimed Baire class. It is therefore the greatest member of $\mathcal{S}_\Theta$.

The graph is Borel as well. By (13), its defining conditions are the coordinate upper bounds and the countably many inequalities

$$\liminf_t q^d(\omega|t) \cdot z \ge \ell^d(\omega).$$

For each d, t, the function $(\omega, z) \mapsto q^d(\omega|t) \cdot z$ is continuous. The displayed comparisons are consequently Borel; no measurable-selection theorem is needed.

If $F \in \mathcal{S}_\Theta$, equivalence (14) and the root bounds give (C), hence value existence by Theorem 2.3. Conversely, if $F \in \mathcal{V}^{[0,1]}_{K,I,J}$, choose a certificate satisfying the root bound for each $d = (p, n)$. The upper boundaries imply $F \le c_\Theta$. For every cluster point $r \in Q^d(\omega)$, the lower boundaries and (13) give

$$r \cdot c_\Theta(\omega) \ge r \cdot F(\cdot, \omega) \ge \ell^d(\omega).$$

Hence (BC) holds, $\Theta \in \mathfrak{T}_{K,I,J}$, and $F \in \mathcal{S}_\Theta$. This proves (11).

To identify the value, fix $F \in \mathcal{S}_\Theta$. Each A^d belongs to $\mathcal{U}(F)$, and so gives the upper bound $v_F(p) \le p \cdot A^d(\varnothing)$ at *every* prior, regardless of the prior assigned to q^d. For rational p, use $d = (p, n)$, (RC) and (3) to obtain

$$p \cdot A^{(p,n)}(\varnothing) \le 1 - B^{(p,n)}(\varnothing) + 1/n \le v_F(p) + 1/n.$$

This proves (12) at rational priors. The right-hand side is concave and 1-Lipschitz, being the infimum of linear functions whose coefficient vectors lie in $[0, 1]^K$. Equation (1) and density

extend equality to all priors. The expression is independent of F, proving the common-value claim.

Finally, pathwise convexity proves the mixing assertion. The weights depend on the public path, with the same weights for all states. If $F \le G \le c_\Theta$, nonnegativity of r preserves every lower inequality in (10); the upper inequalities are already assumed. This proves the last assertion and completes the proof of Theorem 5.1.

6 Strategies and payoff compatibility

The local arrays retain the game's strategic constraints. Their weights reconstruct behavioral strategies as in Section 3. We first make this strategic interpretation explicit, then distinguish these payoff families from ordinary value level sets.

6.1 Fixed-strategy costs

Fix p, F and an uninformed strategy y. Let $\mathcal{A}_y(F)$ be the upper arrays satisfying (UB) and

$$\sum_j y_j(h) A_k(hij) \le A_k(h) \quad (h \in \mathcal{H},\ k \in K,\ i \in I).$$

For an informed strategy x, take the recursive posterior extension (q^x, w^x) in Lemma 3.3, and let $\mathcal{B}_x(F)$ be the $[0,1]$-valued arrays satisfying (LB) for q^x and $\sum_i w_i^x(h) B(hij) \le B(h)$ for every h, j. The two applications of superhedging in Section 3, now with these strategies fixed, give

$$\inf_{A \in \mathcal{A}_y(F)} p \cdot A(\varnothing) = \sup_\sigma \gamma_{p,F}(\sigma, y), \tag{15}$$

$$\inf_{B \in \mathcal{B}_x(F)} B(\varnothing) = 1 - \inf_\tau \gamma_{p,F}(x, \tau). \tag{16}$$

The lower bounds follow from the corresponding supermartingale guarantees; Lemma 3.1 gives arbitrarily close upper bounds. Thus eliminating the capital variables recovers

$$D_F(p) = \inf_{x,y} \left\{ \sup_\sigma \gamma_{p,F}(\sigma, y) - \inf_\tau \gamma_{p,F}(x, \tau) \right\}.$$

This is an identity of infima, with no claim of attainment. The representation does not remove strategy search.

6.2 Common optimal strategies do not determine a family

Example 6.1. Let K be a singleton, $I = J = \{0, 1\}$, and

$$F(\omega) = \mathbf{1}_{\{i_0 = j_0\}}, \qquad G(\omega) = 1 - F(\omega).$$

Both games have value $1/2$, and their optimal strategies are exactly those with a uniform first action; later actions are immaterial. Nevertheless, no $\mathcal{S}_\Theta$ contains both payoffs.

Proof. If both belonged to one family, every array in that family would satisfy the boundaries for both. Since $q \equiv 1$, they would imply

$$\liminf_t A(\omega|t) \ge \max(F, G) = 1, \qquad \liminf_t B(\omega|t) \ge \max(1 - F, 1 - G) = 1.$$

The supermartingale guarantees force $A(\varnothing) = B(\varnothing) = 1$. The certificate cost is then 1, contradicting the root bound for $n = 2$.

Individually, each payoff has a certificate of cost zero. For F, put $A(\varnothing) = B(\varnothing) = 1/2$, and after the first pair keep $A = F$ and $B = 1 - F$. Uniform local weights satisfy the two convex conditions at the root; at later histories the arrays are constant. Interchanging F and G gives the other certificate. □

The same distinction appears in the path-dependent mixture with weights $\lambda_1 = F$ and $\lambda_2 = 1 - F$: it equals 1 everywhere. Its value is 1, whereas mixing within one $\mathcal{S}_\Theta$ would preserve $1/2$. The families therefore record compatibility of pathwise payoff bounds beyond equality of values and optimal strategies.

6.3 A fixed-strategy infimum need not be attained

Example 6.2. Let K, I be singletons, let $J = \{0, 1\}$, and fix the uninformed strategy y to choose each action with probability $1/2$ at every history. Let F indicate the path consisting entirely of zeros. The infimum in (15) is zero and is not attained.

Proof. Identify $I \times J$ with J and write hj for the child reached by action j. The payoff has expectation zero under the fixed fair-coin law. If a nonnegative upper array had root zero, the inequalities $(A(h0) + A(h1))/2 \leq A(h)$ would force every node to have value zero. This violates the boundary on the all-zero path.

For each $N \geq 1$, start instead with $A(\varnothing) = 2^{-N}$. Before time N, double the capital after a zero and set it to zero after a one. After time N, keep it constant. This $[0, 1]$-valued array has the required boundary and satisfies the local fair-coin inequality with equality. Its root values tend to zero. □

The nonattainment in this example concerns the *fixed* strategy y. In the unrestricted certificate problem, cost zero is attained: take $A(\varnothing) = 0$, put $A(h) = 1$ at nonempty all-zero histories and $A(h) = 0$ elsewhere, and set $B \equiv 1$. The upper weights always choose action 1. This satisfies every certificate condition.

Theorem 5.1 concerns the greatest payoff within a *specified* family, not a greatest payoff among all games with the same value. It also supplies neither an optimal strategy nor a zero-cost certificate. The approximation index is retained in the definition of Θ.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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